

// Day 48 of 60

API DESIGN

GraphQL vs REST

→ Swipe through all 6 slides



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The Over-fetching Problem

REST gives you everything

```
GET /api/users/1
```

```
→ returns 50 fields:  
  name, email, age,  
  address_history[],  
  purchase_log[],  
  preferences{},  
  ... etc
```

Frontend needed: name only.

```
// 49 fields wasted
```



Mobile networks suffer the most



The Under-fetching Problem

REST also gives you too little

Need: user + last 5 orders

GET /api/users/1

GET /api/users/1/orders

Two network round trips.

Mobile network = 200ms each

→ 400ms wait time

// One screen, two requests



Waterfalls kill perceived speed



The GraphQL Solution

Ask for exactly what you need

```
POST /graphql
```

```
query {  
  user(id: 1) {  
    name  
    orders(limit: 5) {  
      total  
    }  
  }  
}
```

```
// One endpoint
```

```
// One request
```

```
// Exact shape returned
```



Frontend defines the shape, not backend



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Why REST Is Still King

Standards have power

REST advantages:

- ✓ HTTP verbs (GET/POST/etc)
- ✓ HTTP status codes
- ✓ CDN caching by URL
- ✓ Browser caching free
- ✓ Easier to debug

GraphQL: all POST

→ no URL-based caching



Public APIs almost always stay REST



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The N+1 Problem

GraphQL's hidden cost

Query asks for:

100 users + their posts

Naive resolver:

1 query for users

+ 100 queries for posts

= 101 DB queries

Fix: DataLoader (batching)

→ 2 DB queries total

// Without it: disaster



Always use DataLoader in production



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When to Use Which

Choose by your context

Use GraphQL if:

- Many client types
(iOS, Android, Web, TV)
- Each needs different shape
- Internal tooling

Use REST if:

- Public API
- Simple CRUD
- Heavy caching needs
- Small team



Don't pick GraphQL for the hype



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done = true

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